

THE ARZENS (TAS)

Engineering Internship Program AI, Automation & Security Engineering Track

Assignment 2

Automation Comparison Note: Manual Security Work, Simple Scripts, and SOAR-Style Workflows

1. Manual Security Work

Manual security work refers to cybersecurity tasks that are performed entirely by a human analyst without the assistance of automation tools or scripts. Common examples include reviewing authentication logs line by line, identifying suspicious IP addresses, manually copying information into spreadsheets, preparing incident reports, and monitoring security alerts individually. This approach is most appropriate for small organizations, educational environments, or investigations involving a limited amount of data where human judgment is more valuable than speed.

One of the greatest advantages of manual security work is flexibility. Experienced analysts can identify unusual behavior, investigate complex incidents, and make informed decisions that automated systems may not recognize. Manual analysis is also useful when investigating new or unknown threats that require critical thinking and contextual understanding.

For example, if an organization needs to investigate only ten security log entries after a failed login attempt, manually reviewing each log entry is efficient and allows the analyst to fully understand the incident before taking action.

However, manual work has several limitations. It becomes extremely slow when processing thousands of log entries, is vulnerable to human error, and cannot provide continuous real-time monitoring. As organizations grow, relying solely on manual analysis reduces operational efficiency and increases the likelihood of missing important security events. Therefore, manual security work should not be used for large-scale environments, repetitive daily operations, or situations requiring immediate incident detection and response.

2. Simple Scripts

Simple scripts are small automation programs written using programming languages such as Python, PowerShell, or Bash to perform repetitive security tasks automatically. Unlike manual work, scripts execute predefined instructions consistently and rapidly, making them an excellent solution for tasks that follow predictable rules.

Examples include extracting failed login attempts from log files, filtering suspicious IP addresses, generating CSV reports, checking file integrity, monitoring directories, or sending automated notifications. Instead of manually reviewing thousands of log entries, a Python script can complete the same task within seconds while maintaining consistent accuracy.

The primary advantages of scripting include increased speed, reduced manual effort, improved accuracy, and better productivity. Automation scripts eliminate repetitive tasks, allowing security analysts to dedicate more time to threat hunting, incident investigation, and strategic decision-making. Scripts are also relatively inexpensive to develop and can easily be modified as organizational requirements change.

For example, a Python script can automatically scan a security log containing 10,000 entries, identify all failed login attempts, extract source IP addresses, normalize timestamps, and generate a structured CSV report within a few seconds. Performing the same task manually would require several hours and significantly increase the possibility of human error.

Despite these benefits, scripts also have limitations. They perform only the tasks they were designed to accomplish and may fail if log formats change unexpectedly or if proper error handling has not been implemented. Poorly written scripts may introduce security vulnerabilities or produce inaccurate results. Therefore, developers should always validate input data, implement exception handling, document their code, and test scripts thoroughly before deployment. Simple scripts should not be relied upon when multiple security tools, approval workflows, or complex decision-making processes are required.

3. SOAR-Style Workflows

Security Orchestration, Automation, and Response (SOAR) platforms represent the most advanced stage of security automation. Unlike simple scripts, SOAR platforms integrate multiple security technologies into a unified workflow capable of collecting alerts, enriching threat intelligence, creating incident tickets, notifying analysts, executing predefined response actions, and maintaining complete audit trails.

SOAR platforms are most appropriate for medium and large organizations that receive thousands of security events every day from firewalls, intrusion detection systems, endpoint protection platforms, email gateways, and cloud security services. Instead of requiring analysts to investigate every alert manually, SOAR systems automatically prioritize incidents, reduce false positives, and coordinate responses across multiple security tools.

For example, when a malicious IP address attempts repeated unauthorized access, a SOAR platform can automatically collect the related logs, verify the threat using external intelligence feeds, create a helpdesk ticket, notify the security team, isolate the affected endpoint, and block the malicious IP address without requiring continuous manual intervention.

The major advantages of SOAR include rapid incident response, improved operational efficiency, centralized management, standardized workflows, regulatory compliance, and comprehensive audit logging. These capabilities allow organizations to respond to cyber threats much faster while reducing analyst workload.

However, SOAR platforms are expensive to implement and require experienced personnel for configuration, maintenance, and continuous optimization. Incorrect workflow design may automate inappropriate actions or disrupt normal business operations. Therefore, SOAR solutions should not be implemented in very small organizations or for simple security tasks where the complexity and cost outweigh the operational benefits.

Conclusion

Security automation should be adopted gradually according to the organization's size, operational requirements, and security maturity. Manual security work remains valuable for education, detailed investigations, and situations where human judgment is essential. As workloads become repetitive and larger datasets must be processed efficiently, simple automation scripts provide a practical and cost-effective solution that improves productivity while reducing errors. When organizations require continuous monitoring, rapid incident response, integration across multiple security technologies, and enterprise-scale operations, SOAR platforms become the most appropriate solution.

Ultimately, cybersecurity professionals should view automation as a tool that enhances human expertise rather than replacing it. A balanced combination of skilled analysts, reliable automation scripts, and enterprise orchestration platforms creates a secure, efficient, and scalable cybersecurity environment capable of addressing modern security challenges.

AI Assistance Note

This assignment was prepared with the assistance of ChatGPT (OpenAI) for brainstorming ideas, improving technical writing, refining document structure, and enhancing readability. The final document was carefully reviewed, understood, and edited by the author before submission. The author accepts full responsibility for the accuracy, originality, and final content of this assignment.